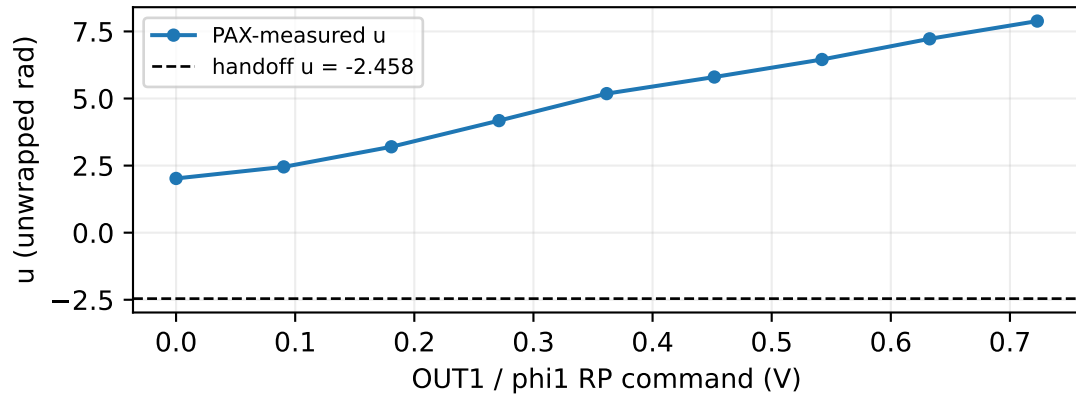
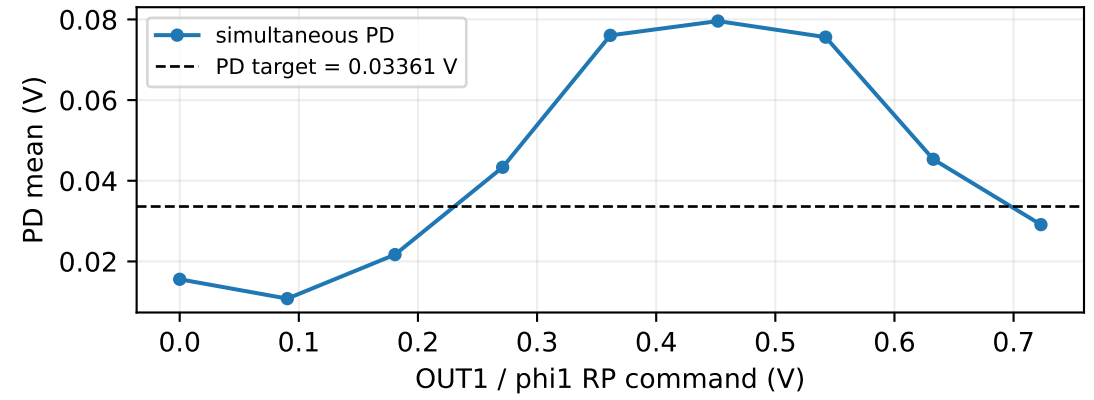


Phi1 PD FPGA lock — 005716_phi1-pd-lock-test_phi1-pd-lock-4

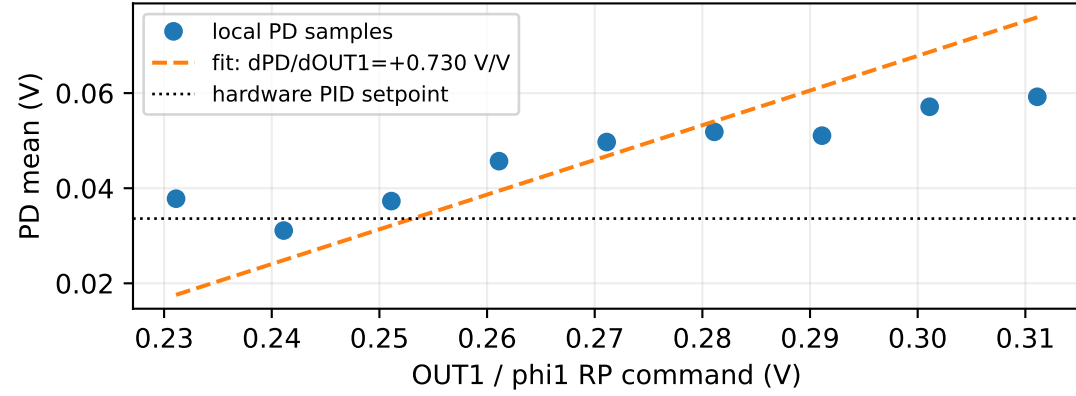
PAX-assisted ϕ_1 rough map



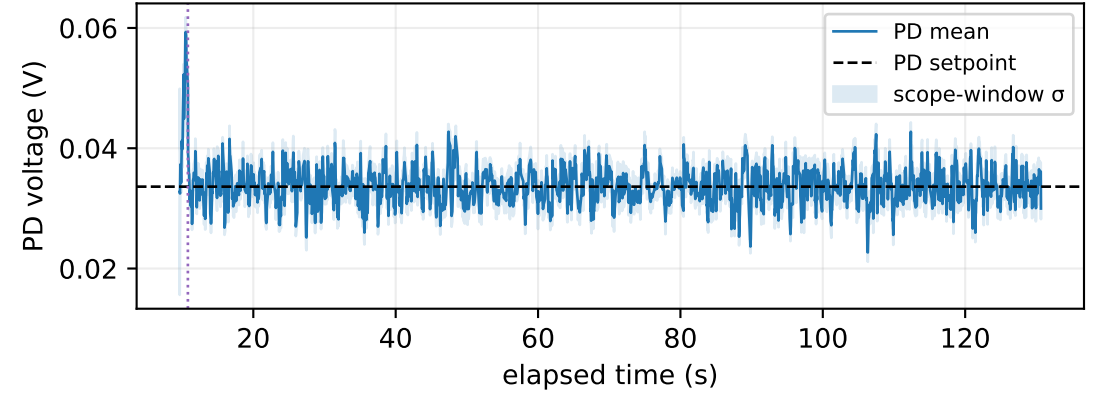
PD value associated with rough u map



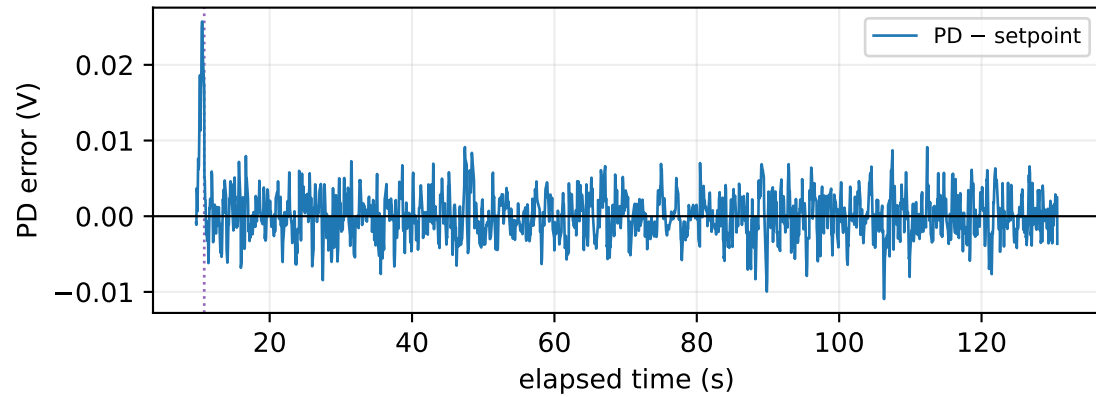
Automatic local PD calibration



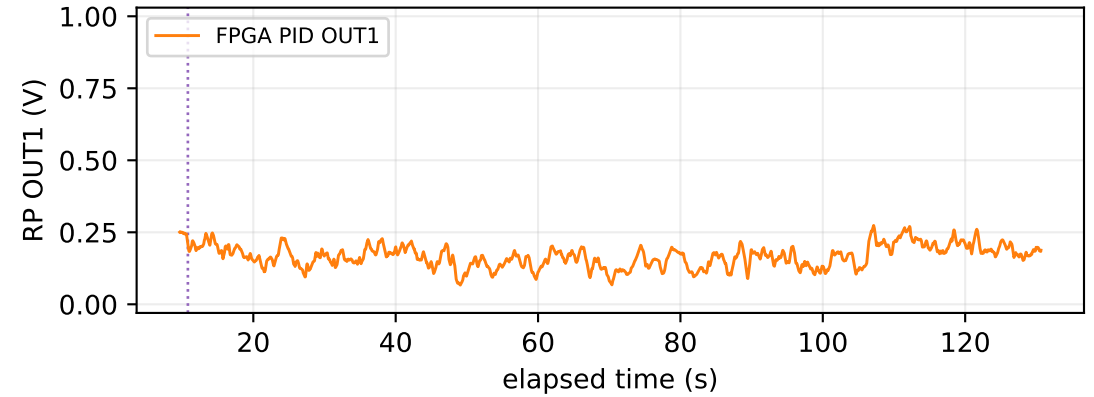
FPGA PD lock



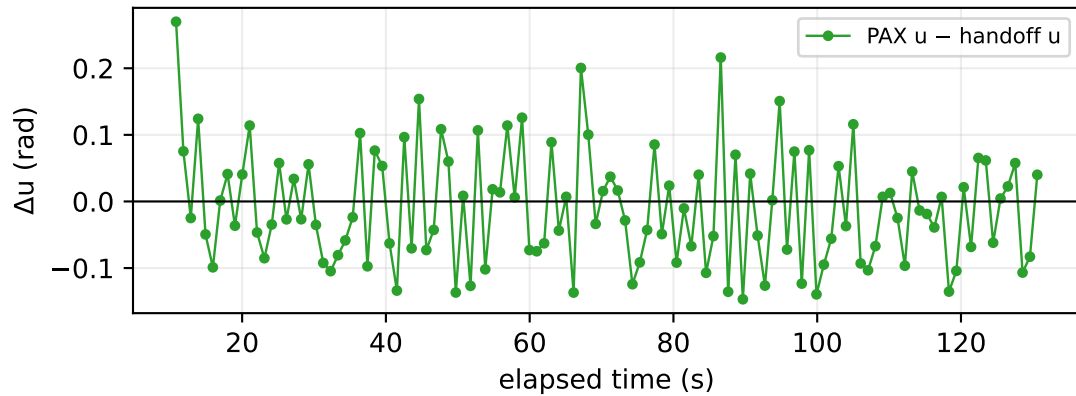
PD locking error



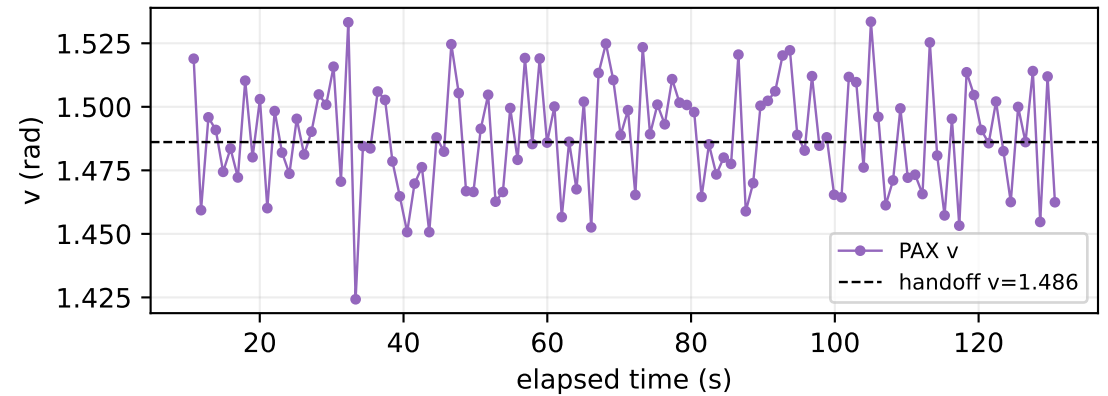
Continuous FPGA actuator output



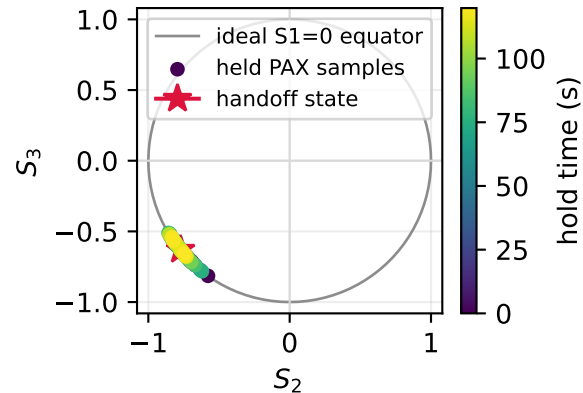
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



Interpretation

- Tight cluster near the red star: PD lock preserves the selected u branch.
- Motion around the circle: residual ϕ_1/u drift or branch movement.
- Radial motion: polarization is leaving the ideal equator.

PAX handoff target $u = -2.4575$ rad PD setpoint = 0.03361 V local slope = +0.7296 V/V
 FPGA PID: P=-0.302, I=-5 Hz Duration=121.0 s median |PD error|=0.00188 V (late 0.00185 V) OUT1 range=0.067...0.274 V PAX median $|\Delta u| = 0.064$ rad (118 snapshots)